MATS 12 · NEEL NANDA STREAM · EXPLORATION PHASE · 30 OCTOBER 2026

When do meta-models reveal more than the prompt?

Reporting gaps and read-site limits in Oracle Lens and NLA

Qwen3.6-27B · synthetic banks and one public benchmark

Guiv Farmanfarmaian · **Suraj Prasad**

Our main claim, two supporting claims, and how much we believe them

75% · 45%

Main claim · readouts are mostly the prompt, rebuilt; Qwen's own state shows only where it is large at the read token

75% rebuilt prompt. Down if a reader tracks verified model-private state and beats a text-only baseline

45% own state, late. Down if restoration works at block 44 on a fresh unsteered bank; up if chains follow the holder, not the last hand-over

85%

Supports it · A: both readers carry structural state in their words, and almost never state it

Down if: human labels disagree with the judge. Copying the last 32 tokens was tested and mostly ruled out (exp 31)

Up if: it holds on a second model

55%

Explains the NLA · B: its silence is a warm-start habit the reward neither pays for nor removes

Test: RL against a critic that pays for statements

Up if naming appears; down if silence stays

Credences set before the talk. Main claim: exp 06, E100, E126, E135, E137, E140. Claims A and B: Suraj's sprint (exps 22–30).

What existed before the sprint, and what the sprint kept

Suraj, before 19 Oct

- Readers replicated: FVE 0.761; WorkspaceBench 12 of 12 inside our CIs
- J-space survival test can't separate in-J from non-J
- Confirmatory run pre-registered (rev 4.3), never run

Guiv, before 19 Oct

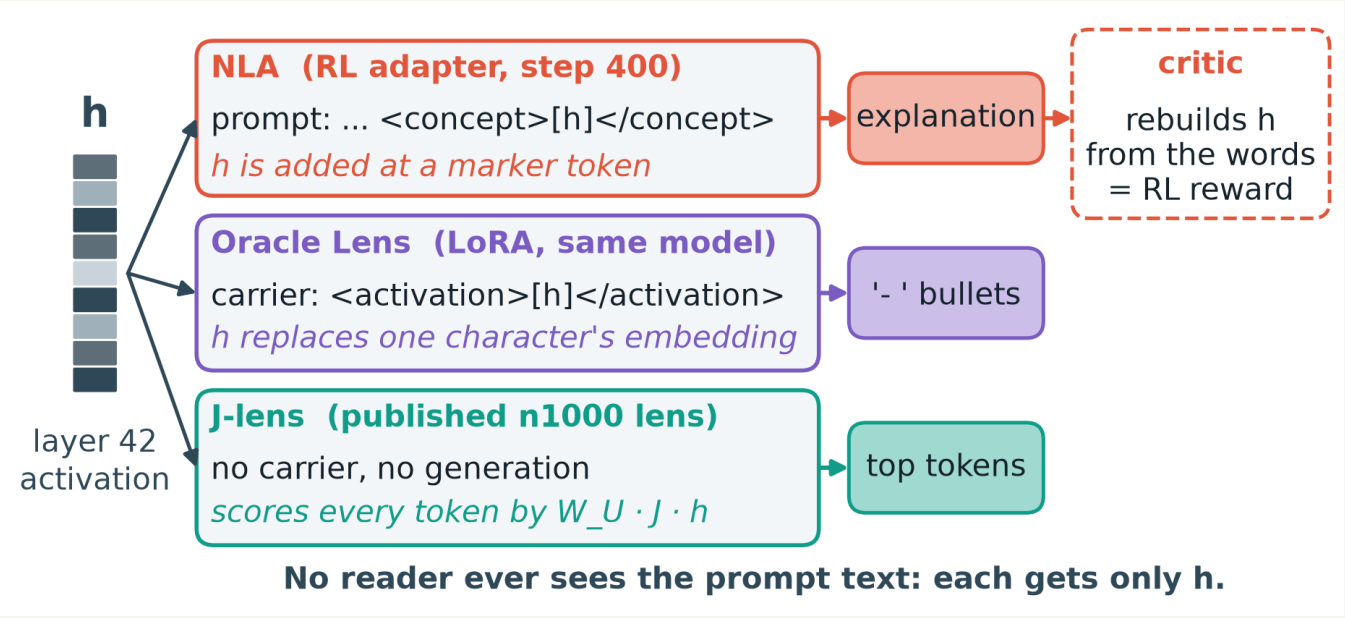
- Question: when can we trust a reader? Does agreement mean it's right?
- 297 puzzles the model got wrong: J-lens and Oracle Lens recover its answer reliably only around block 52; a probe at 42
- With 29 candidate names instead of 2: about 2–6% correct
- Non-J part flips 64/64 answers, J part 0/64, yet Oracle prefers the planted name 53% vs 70%

What the sprint kept

- Replication and the WorkspaceBench re-solving result
- Agreement → E94: precision 0.94 but only 23% of items; so, what do readouts reveal beyond the prompt?
- Late reads, sensitivity ≠ causal importance → the causal tests (Part 3)
- 23 Oct: Neel asked what one method finds that another can't

Suraj: deck v6 (5–13 Oct), notes/decisions.md (D35) · Guiv: pre-sprint project proposal to Neel; E94 in guivff/activation-readers @ c60fa970

Setup: one model, three readers, and nobody sees the prompt



What to keep in mind

- Qwen3.6-27B: 64 blocks, width 5120
- Oracle Lens: LoRA; the activation replaces one character's embedding; trained at blocks 20–60, step 4
- NLA: writes an explanation; a critic rebuilds the activation from it; RL pays for the rebuild
- Reads at block 42 (NLA) and 44 (Oracle)
- A blind LLM judge decides what a readout states

Three questions about every state

Decodable

- A linear probe can read it
- Held-out AUROC

Causal here

- Qwen's answer depends on it at this token and block
- Patch it back; normalised effect R (0 = none, 1 = full)

Stated

- The reader's text says it
- Blind judge; k of n, missing shown

The three come apart all the time. Most of this talk is about where.

What readers write: often a prompt of this kind, rebuilt and answered

Prompt: “Elif carries the rope and Jonas carries the lantern. Who carries the lantern? Name only.”

Oracle Lens: “- Alice / - John …”

NLA: “… must provide just the name “Alex” as the direct answer to “Who sits first.” …”

Neither names Jonas or Elif. Both rebuild a name-only question and answer that.

Prompt: “What is the opposite of 'hot'? (Answer in Spanish, one word.)”

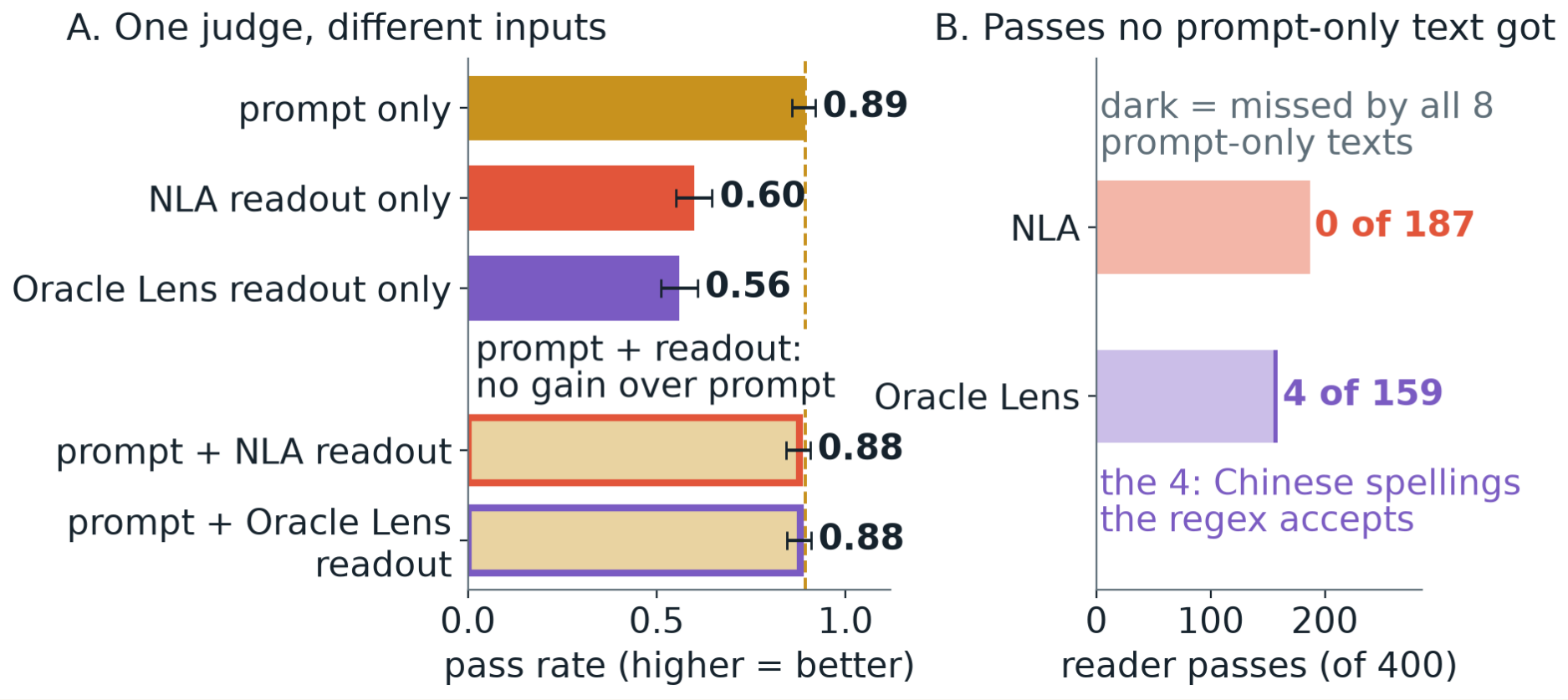
Oracle Lens: “- Frío / - Frío …”

NLA: “… now delivering a Spanish-language direct answer …”

When the rebuild keeps the specifics, the answer is right.

Verbatim, shortened with … · G1 pilot items lang-02A, own-02A · block 42, last prompt token · guivff/activation-readers @ c60fa970

Negative result 1: the benchmark can't tell reading from re-solving the prompt



- Prompt alone passes 0.89; adding either readout adds nothing (NLA −0.015 [−0.043, +0.013])

Lesson: to ask what a reader captures, hold the text fixed and move the activation

Suraj exps 01, 06 · Jarus77/mats-nla-workspace @ 4d43ecb2 · text arms are best-of-8 against k = 1 readers

When Qwen's answer changes with the wording, the reader rarely follows

The test

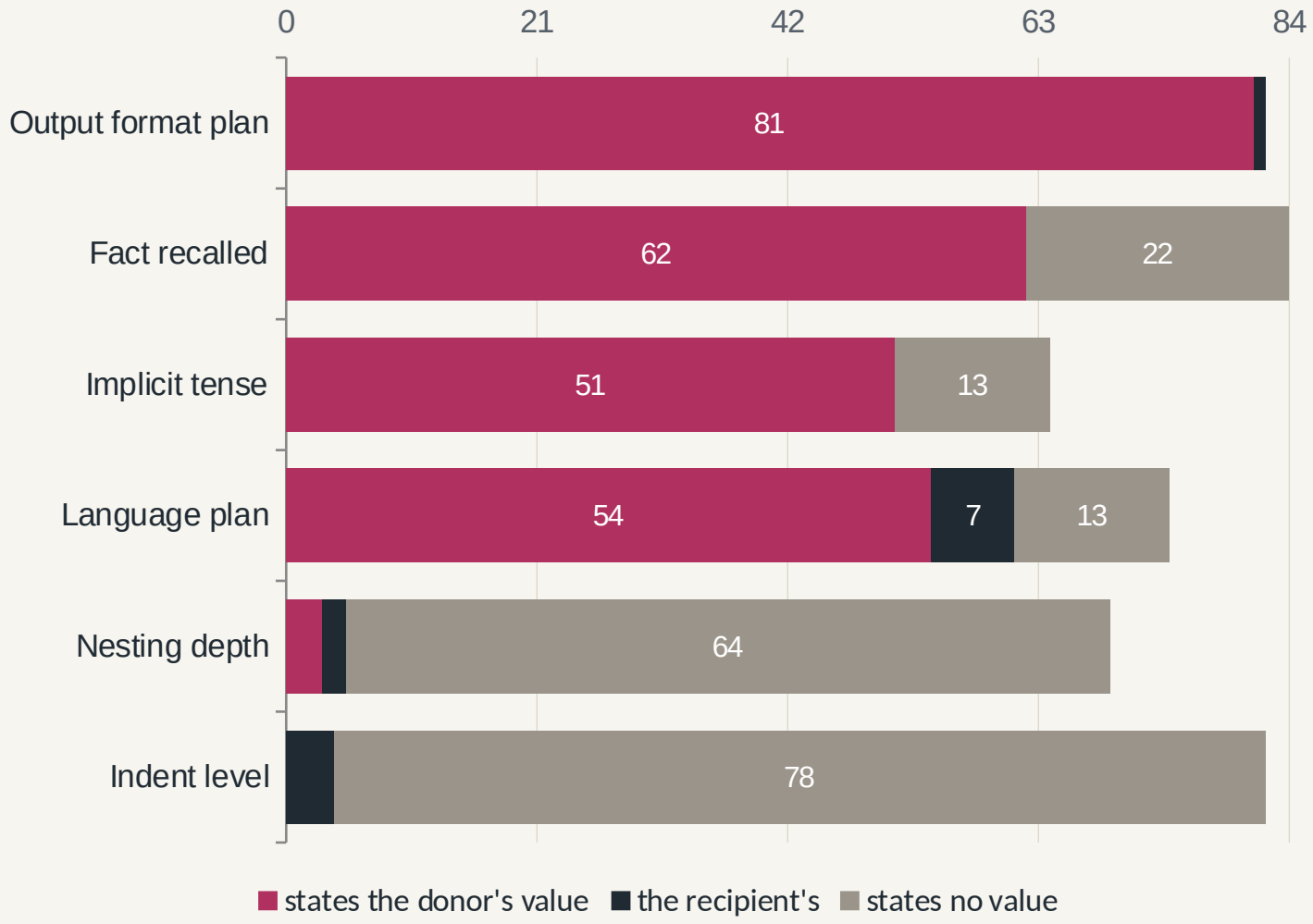
- Same fact, two wordings: “What is the capital city of X?” / “Which city is X's capital?”
- Keep pairs where Qwen answers differently: 58 of 300
- Read both at block 60. Does the reader follow Qwen?

Result (12 evaluable pairs)

1 of 12 follows Qwen · 0.08 [0.01, 0.35]

- 11 of 12 give one answer for both wordings
- Caveat: only 12 of 58 pairs were evaluable

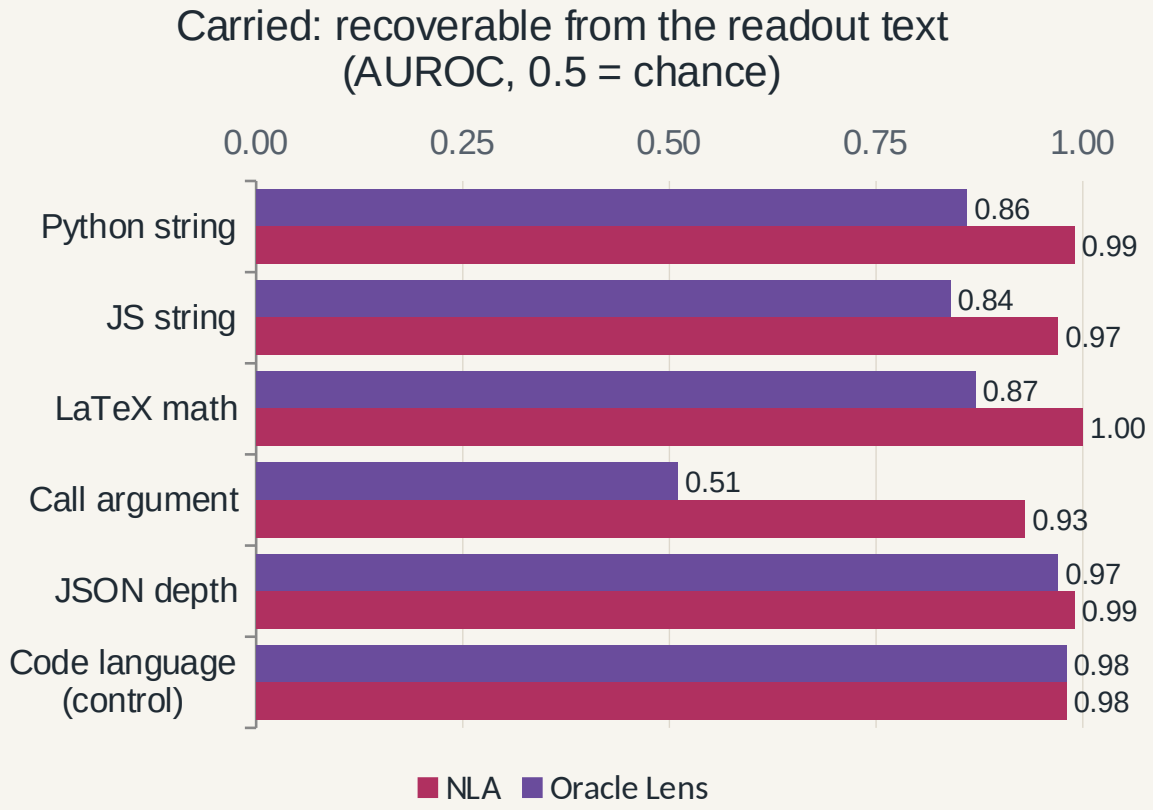
Swap in another item's activation: content follows it, structure goes silent

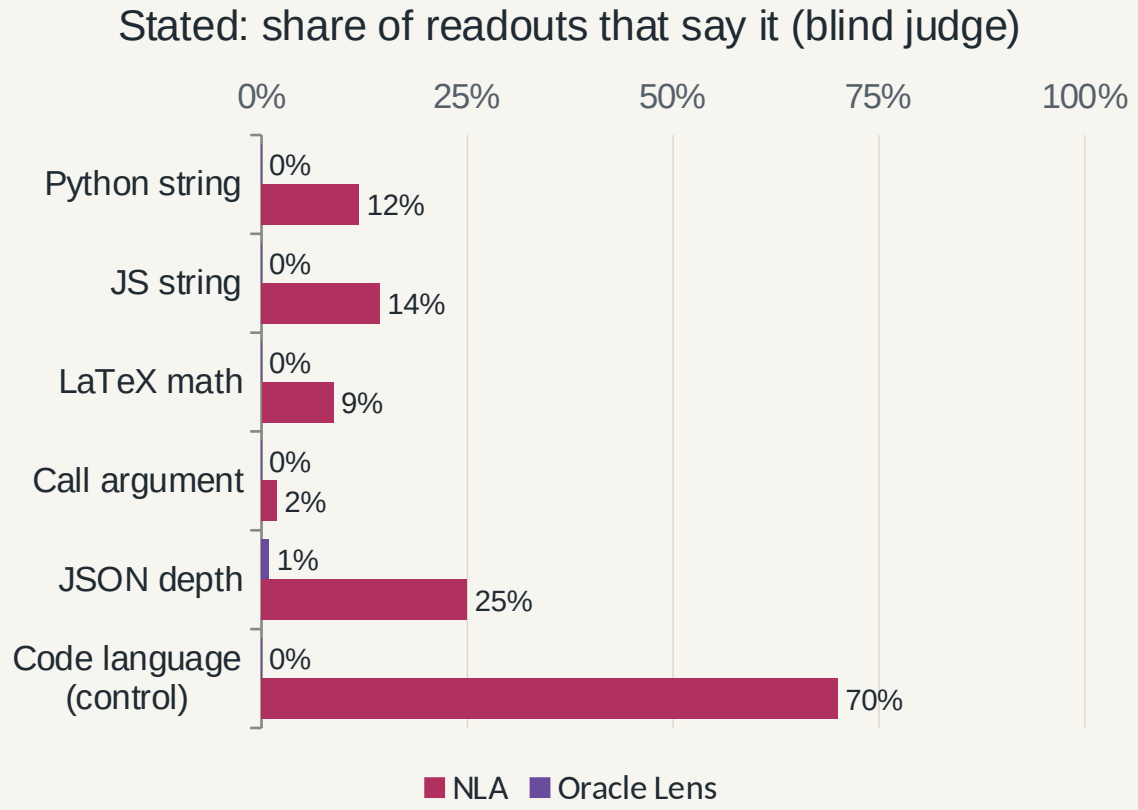


- Content (format, fact, tense, language) follows the donor
- Nesting and indent: the readout states no value at all (64 and 78 of 84)
- Other categories (both, exhibits only, missing) not drawn

E100 (confirmatory, R-PC MET 5/5) · Oracle Lens, block 44, 84 items per row · guivff/activation-readers @ c60fa970

Claim A: both readers carry structural state in their words and almost never state it



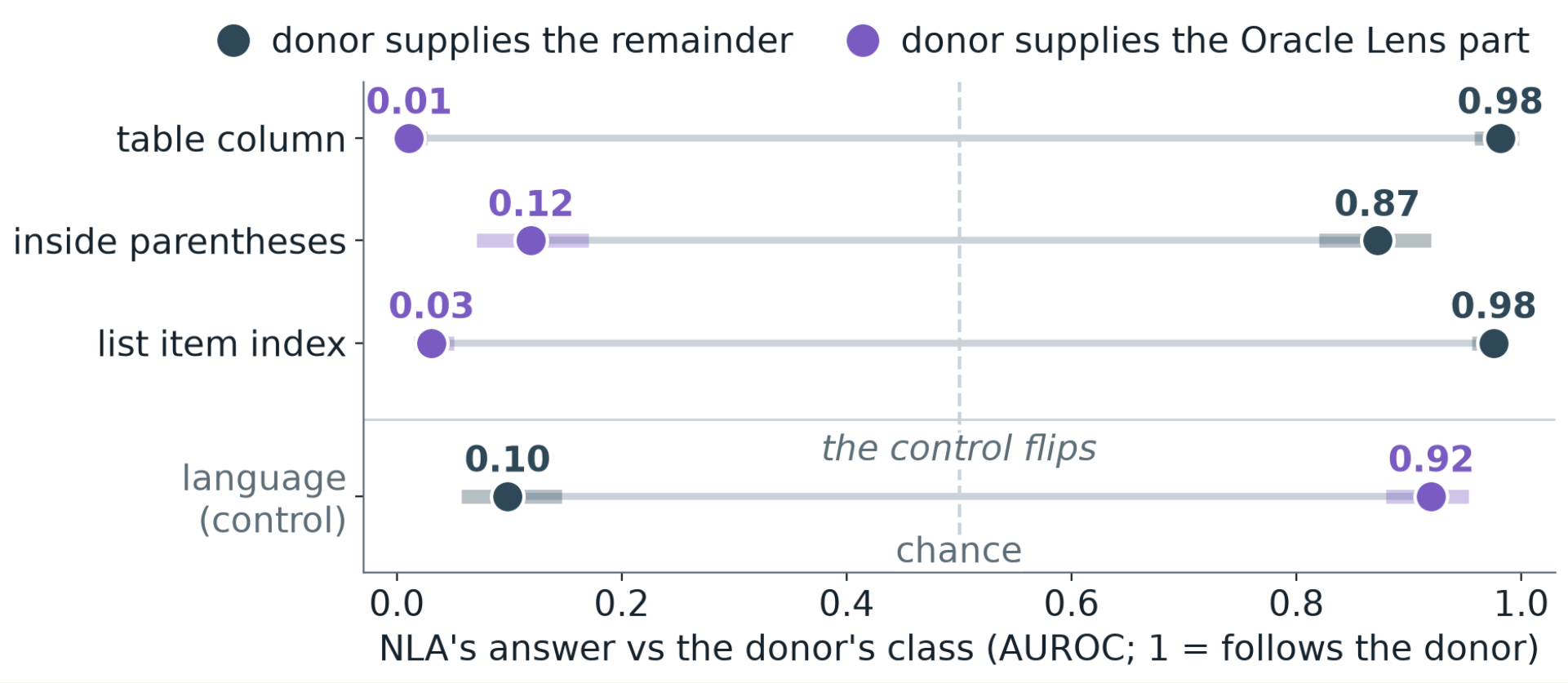


- Structure is recoverable from both readers' words (0.84–1.00 except Oracle Lens on call argument); Oracle Lens states it 0–1%, the NLA 2–25%

Caveat: the critic is the NLA's own, so Oracle Lens's text is rebuilt out of distribution; Markdown rows are not linearly present and are left out

Suraj exp 25 (carried: critic rebuild, frozen probe, 160 items per row) and exp 22 (stated: blind judge, 40 items × 8) · Jarus77/mats-nla-workspace @ 4d43ecb2

The NLA reads position from the part of the vector Oracle Lens's words do not rebuild

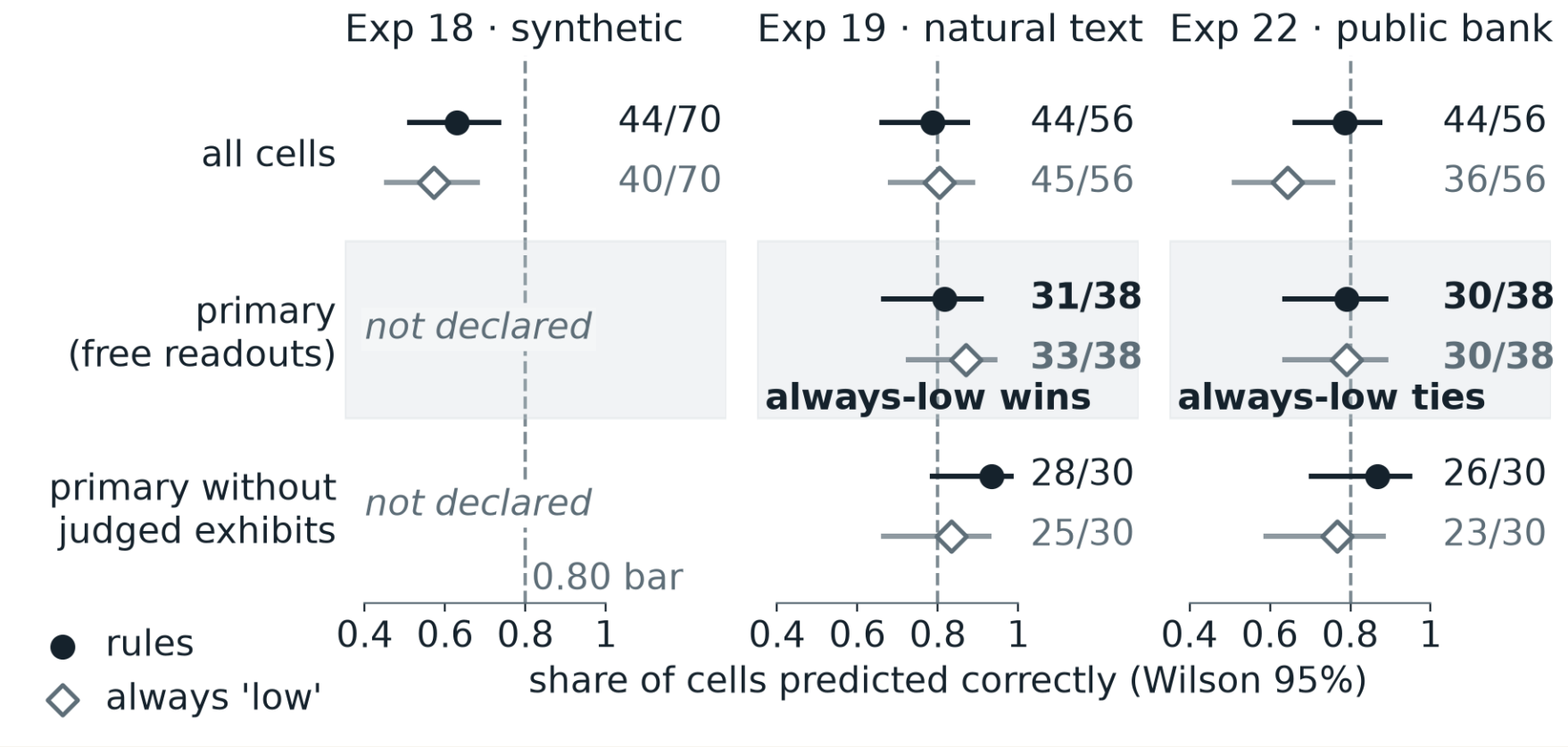


- Table column: follows the donor via the remainder 0.98, via the Oracle Lens part 0.01

Unreconciled: Guiv's E7 swap gives 0.39 on the same row; E27 could not assign the gap to one cause

Suraj exp 16 (S16: Suraj's design, run by Guiv's agent on Suraj's pipeline), 200 items per row · Jarus77/mats-nla-workspace @ 4d43ecb2 · E7, E27: guivff/activation-readers @ c60fa970

Negative result 2: our prediction rules tie or lose to always predicting 'low'



- Primary scorecards: rules 31/38 vs always-low 33/38 (exp 19); 30/38 vs 30/38 (exp 22)

Lesson: pre-register the baseline too. Found in the 30 Oct raw recompute, after we had quoted these as passing

Claim B, at 55%: a warm-start habit, not the reward, keeps the NLA from naming structure

Not the reward (exp 29)

- Over 800 RL steps, reconstruction rises (FVE 0.18 → 0.59)
- Naming the code language rises (0.55 → 0.71–0.77)

Naming structure stays flat: 0.00–0.29, no trend

Not capacity (exp 30)

- A 400-step adapter learns to name it
- At no cost: FVE 0.507 vs control 0.512
- Correct 0.79–1.00 on held-out files; but a held-out row only 0.53

Weakest link

- A replay of one training run, with no counterfactual
- RL against a critic that pays for statements was never run: that is the test

The NLA can name structure at no cost, yet RL never makes it; so the silence looks inherited from the warm start, not paid for.

Suraj exps 29 (RL replay, one final critic scores every checkpoint) and 30 (400-step naming adapter) · Jarus77/mats-nla-workspace @ 4d43ecb2

Causal test: push Qwen's answer, then put the read token back

1 · Push (E104)

- At block 40, add a direction toward the other answer on the 8 tokens before the read token
- Qwen's answer flips: 84 / 84 on the fact row

2 · Restore (E126)

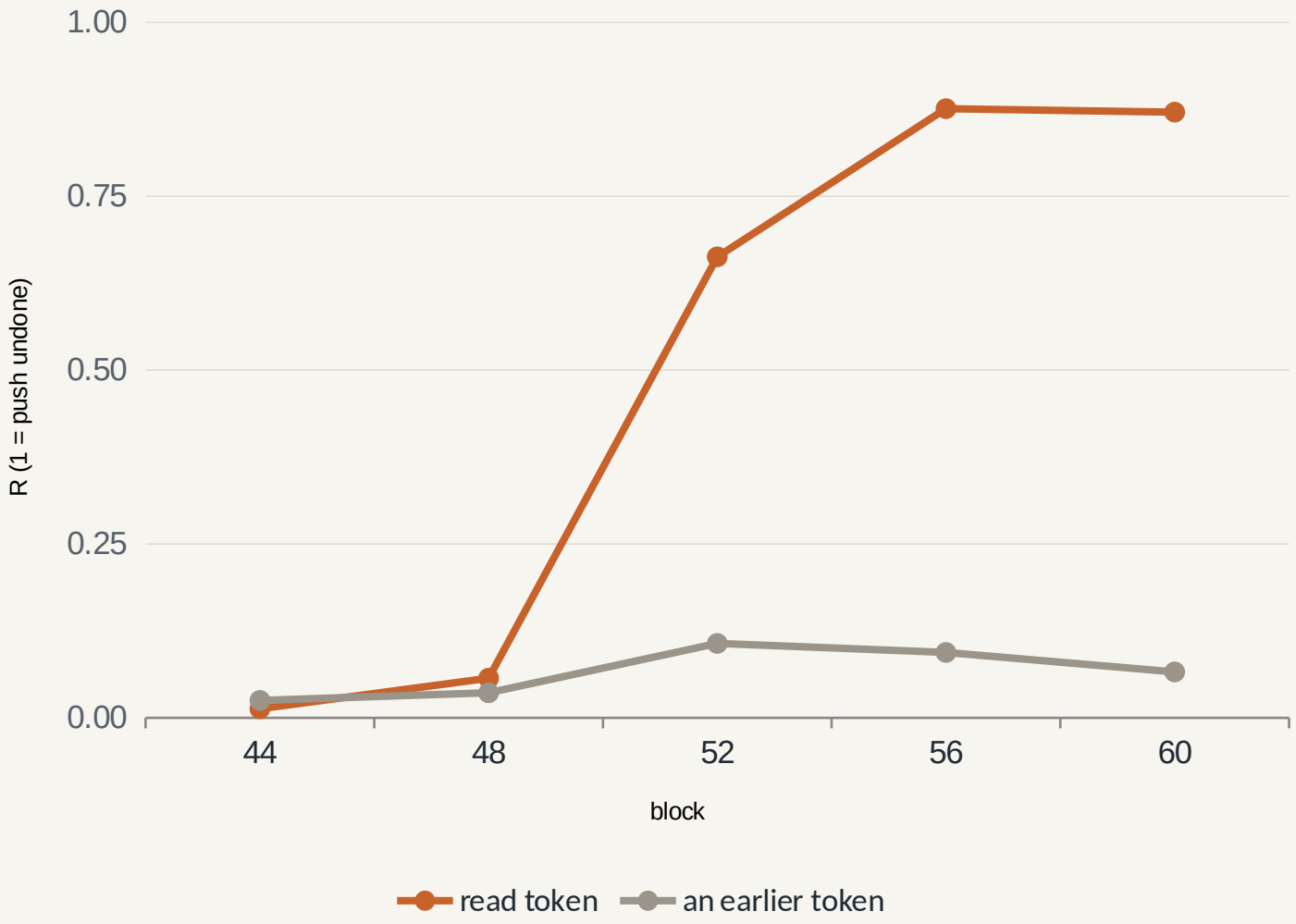
- At block L, copy back the clean activation at the read token only
- If Qwen's answer comes back, the read token at L carries it

3 · Score

- R = 0: restoring does nothing
- R = 1: fully undoes the push
- Controls: an earlier token, a random direction

E104 steering, E126 restoration · guivff/activation-readers @ c60fa970 · d: class-mean difference of block-40 activations on 84 dev items · at α = 8 the push is 2.23× the median block-40 norm

Restoring the read token works late: between blocks 48 and 52

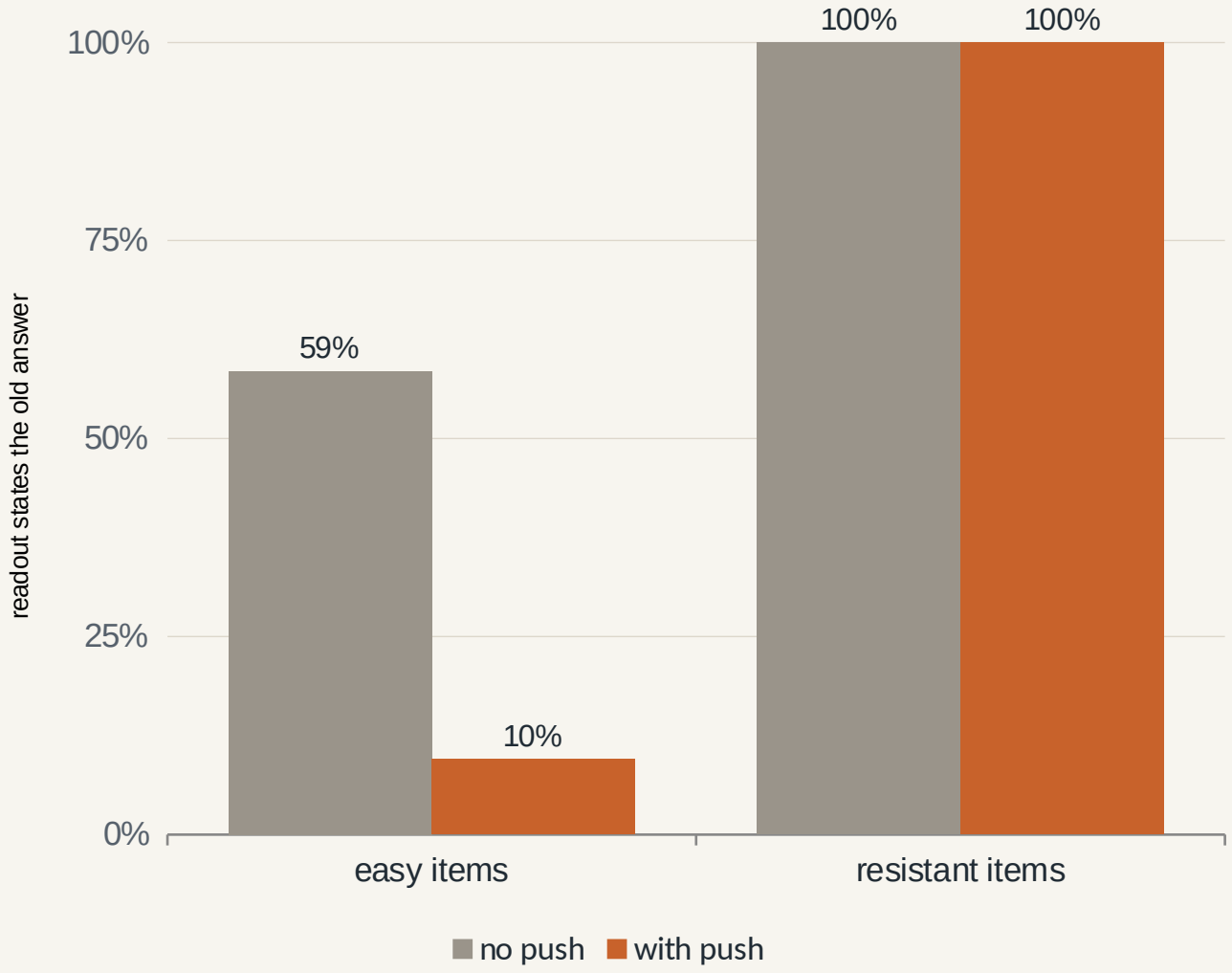


- R ≈ 0 at 44–48, 0.66 [0.51, 0.80] at 52, 0.87 at 60
- Controls stay near 0: an earlier token ≤ 0.11; a random direction |R| ≤ 0.01
- Oracle readout: still the old answer at 44; the new one on 84/84 at 52 and 60

Caveat: one fact pair (France vs Peru) in 84 contexts, under a 2.2× push

E126 (R-MED1 MET) · fact row, 84 items, median R · guivff/activation-readers @ c60fa970

Early 'stale' readouts track Qwen's own unpushed answer



- Resistant items: the old answer is in the readout even without the push (42/42)
- Easy items: the push changes the readout (24/41 → 4/42)

Caveat: groups defined after the fact; association only

E135 (α 6) and §248C (clean run, exploratory) · guivff/activation-readers @ c60fa970

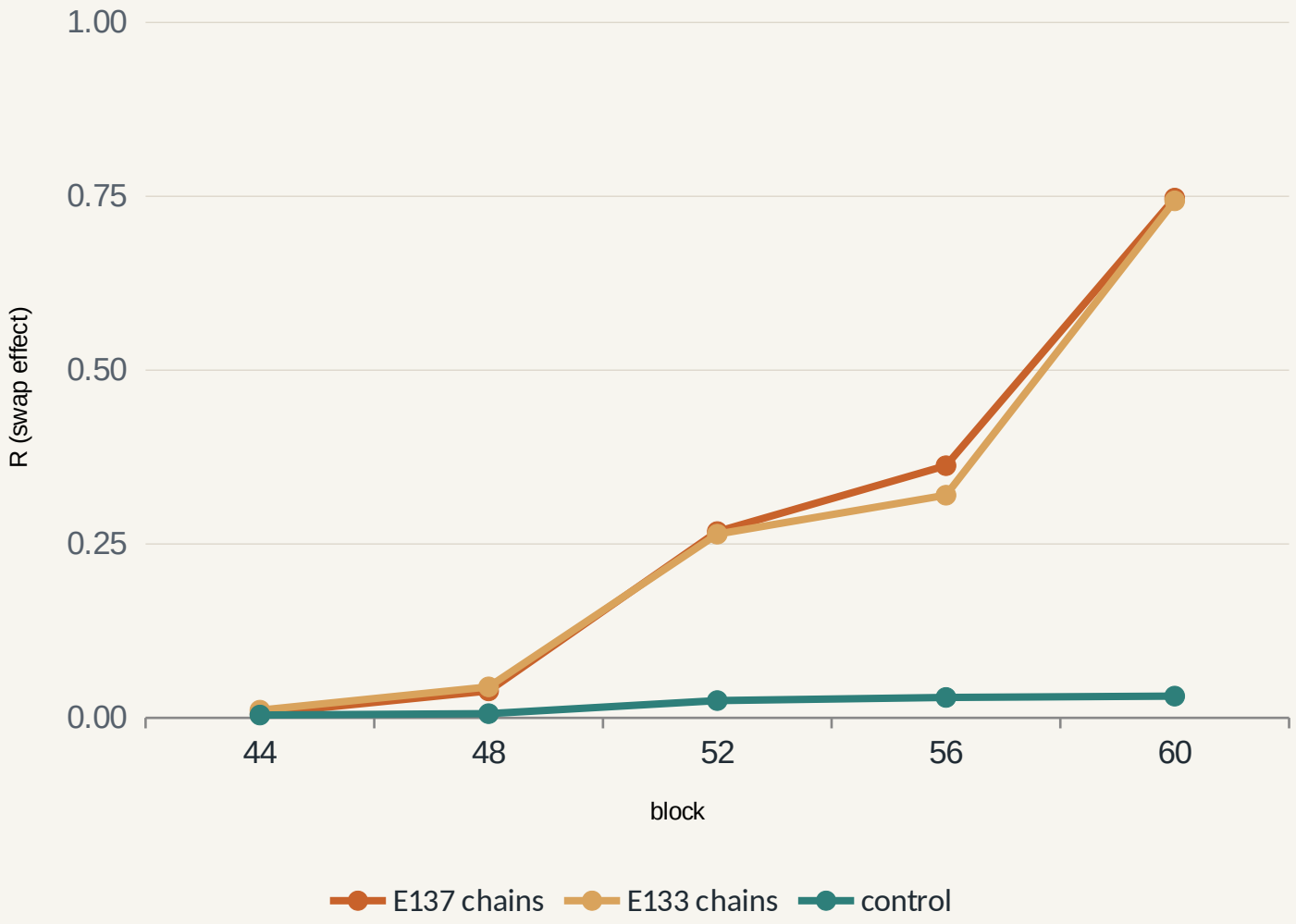
Without any push: who holds the key? Twin chains differ in one name

```
Twin A: The brass key starts with Alice. Alice hands the brass key to Bob. Bob hands the brass key to Dana. Dana hands the brass key to Erin. Later, Carl asks Dana about the brass key. Bob waves to Carl. The brass key is now held by
```

```
Twin B: … Bob hands the brass key to Dana. Dana hands the brass key to Frank. Later, Carl asks Dana about the brass key. Bob waves to Carl. The brass key is now held by
```

- Swap twin B's activation into twin A at the read token (“ by”), block by block; R = how far Qwen's answer moves
- Anti-recency: names after the last hand-over, so the most recent name is never the holder

The final token matters mostly at block 60, and the previous holder is rarely named



- R 0.27 at 52, 0.75 at 60; the registered rule (R ≥ 0.50 at 52 or 56) was NOT MET
- Previous holder named in ≤ 4% of readouts

Open: holder state vs reading off the last hand-over are not yet separated

E137 (R-NL2 NOT MET), E133 · guivff/activation-readers @ c60fa970

How we kept LLM-driven research honest, and where it slipped

Guardrails

- Banks, instruments and predictions hashed before any reader ran
- Separate agent sessions re-ran analyses on sealed outputs
- Falsifiers that fired: exp 18 missed its bar; the 0.30 rule withdrawn

Where it slipped

- Raw recompute (30 Oct): 49 written numbers disagreed with raw files; two changed a conclusion
- A swap control was invalid by construction; an agent reviewer found it (E138 → E138R)
- Judge vs one human coder: κ 0.68, under our 0.70 bar

Suraj: D49, D51 · Guiv: E138R, E71, E27 · reviewers R-2…R-5, R-Q are agent sessions, not human replications

Every headline number has a baseline beside it

Baseline	Rules out	Result
Prompt-only text model	Re-solving the prompt	0.89 vs readouts 0.60 / 0.56
Always predict 'low'	Scorecard padding	Ties or beats the rules on primary cells
Shuffled 32-token bag	State is just vocabulary	Structure falls to 0.44–0.50
20 random directions	Push size alone	Statement direction moves the model z +0.7
Earlier token, random direction	Any patch restores	R ≤ 0.11 at every block (E126)
Control chains	Any swap moves the answer	R ≤ 0.031 at every block (E137)

Suraj exps 06, 15, 19, 20, 22 · Guiv E126, E137 · Jarus77/mats-nla-workspace @ 4d43ecb2 · guivff/activation-readers @ c60fa970

Ways this could be boring

Boring explanation	Status	What we have
Readouts just copy the last 32 tokens	Mostly no	Exp 31: readers reuse window words at 3–6× chance, but deleting them leaves NLA carrying unchanged and costs Oracle Lens 0.01–0.08
The judge is wrong	**Partly**	κ 0.68 vs one human; Suraj's judge has no human labels
One model, one NLA run	Open	Every claim is Qwen3.6-27B only
The fact row is one fact	True	France vs Peru in 84 contexts
The chain effect is recency	**Partly**	Names after the hand-over ruled out; last hand-over not
The warm-start story is a replay	**Partly**	Flat over 800 steps; no counterfactual run

Suraj exps 20, 26, 29 · exp 31 (text-bag measure; critic not re-run): Jarus77/mats-nla-workspace @ 10d495f1 · Guiv E56, E71, E137 · guivff/activation-readers @ c60fa970

Where the claims stand

Registered, and met

- Swapped-in activation followed for content (E100, 5 of 5 rows)
- Restoration works late, not at 44 (E126)
- Paraphrase pairs: one answer for both (E140, 12 pairs)

Consistent, read afterwards

- Text-only parity on the benchmark (exp 06)
- Claim A: carried, rarely stated (exps 22–25)
- Unsteered chains causal late (E137)
- Claim B: naming flat over RL (exp 29)

Open: would change our minds

- A reader that tracks verified private state and beats a text-only baseline
- Restoration R ≥ 0.5 at block 44 on a fresh unsteered bank
- A statement-aware critic that makes the NLA name structure

Registration statuses: results/g_agreement/writeup.md · guivff/activation-readers @ c60fa970

In hindsight, what we would do from day one

1 Pre-register the baseline

Always-low would have flagged the scorecards on 27 Oct.

2 Audit against raw files weekly

The 30 Oct recompute found 49 mismatches.

3 Read the judge's actual inputs

The E138 artefact sat in plain sight.

4 Count distinct items, not prompts

84 fact prompts were one fact pair.

5 Build the decisive test early

A statement-paying critic; holder vs last hand-over.

6 A second model early

Every claim is still one model.

Suraj: D51, exp 18 · Guiv: E138R, E56 bank

What we learned

About the readers

- Decodable, causal here and stated come apart; check each separately
- A readout is mostly a rebuilt prompt; text-only baselines are hard to beat
- A reconstruction reward does not pay for naming; whether a reader names a state looks like training habit more than capacity

About doing research

- Know what a benchmark holds fixed before trusting its score
- Pre-register the baseline along with the prediction
- Agent-written research drifts from its data; audit against raw files, read the inputs, count items

Next, in order of cost

Experiment	Question	Cost
Judge vs human labels, 600 readouts	Are naming rates real?	No GPU, ~10 rater-h
Chains where the last hand-over is undone	Holder state or last hand-over?	[estimate]
Fact row on many fact pairs	Does late restoration generalise?	[estimate]
Second model: Llama-3.3-70B NLA	One run, one model	~$30–60, 1–2 days
Benchmark of verified model-private state	Does any reader beat text-only?	[estimate]
Statement-aware critic, then RL	Claim B directly	~$500–800, 3–5 days

Costs: Suraj's estimates for his rows; ours not yet costed · Jarus77/mats-nla-workspace @ 4d43ecb2

CLOSING

Decodable, causal here, and stated are different.

A reader faithful to the text is not yet faithful to the model.

Both readers carry structural state and almost never say it; they report Qwen's own state mainly where it is large at the read token, and late.

APPENDIX

Backup figures

Suraj's per-experiment figures; the detailed 39-slide deck has every Guiv experiment with full captions.

Each concept is measured on the activation alone, with lines fixed before any reader ran

On the activation alone

readable **AUROC ≥ 0.80**
linear probe on h

span-sayable **AUROC ≥ 0.80**
critic rebuilds h from real text

statement shift **≥ 0.30**
critic on a sentence (withdrawn)

used **|z| ≥ 3**
push h; the next token moves

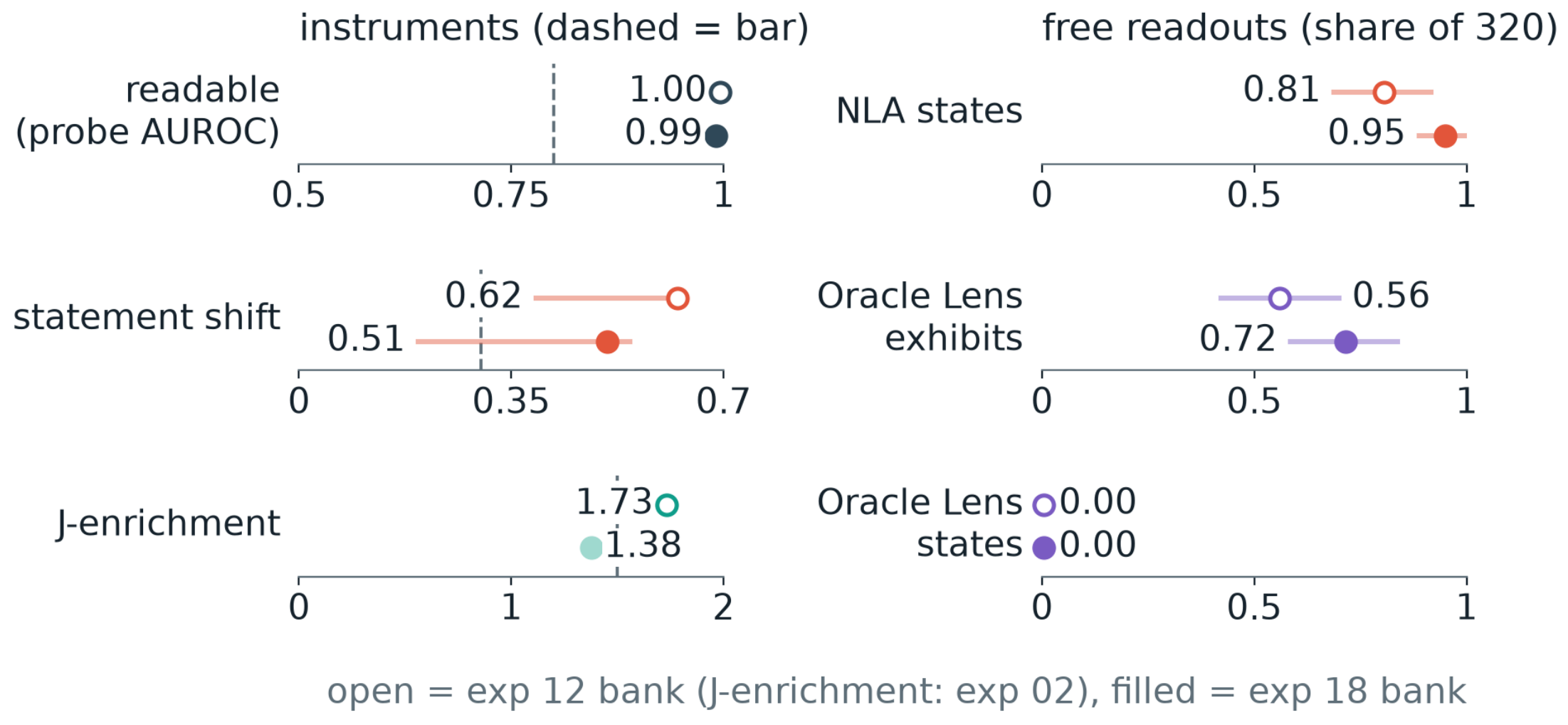
J-enriched **≥ 1.5**
direction inside the J-space

predict →

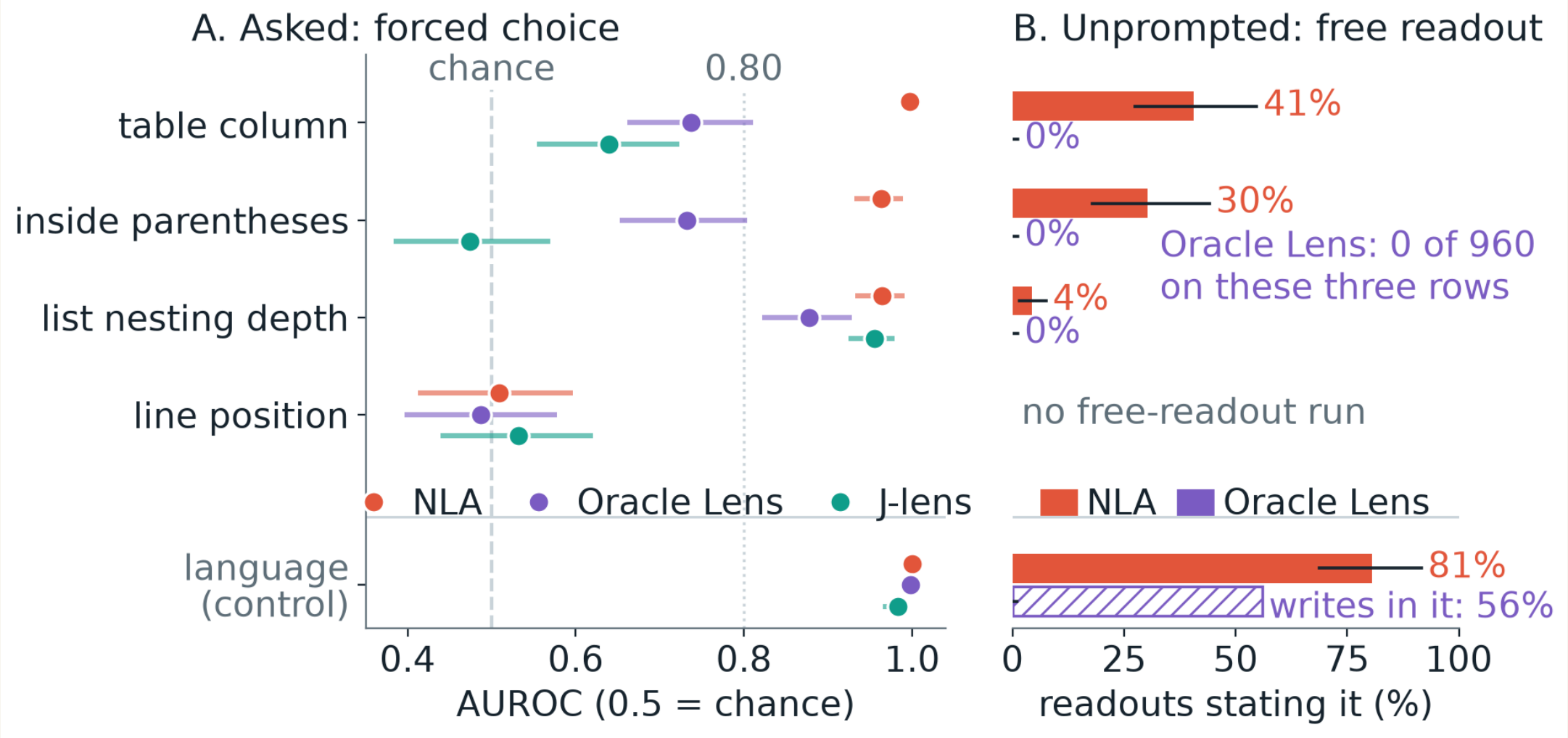
Reader scores (pre-registered lines)

	NLA	Oracle Lens	J-lens
prompted (AUROC)	≥ 0.85	≥ 0.70	≥ 0.70
states (judge)	≥ 20%	≥ 5%	–
exhibits (judge)	–	≥ 30%	–
complement (multi-choice)	≥ 20%	–	–

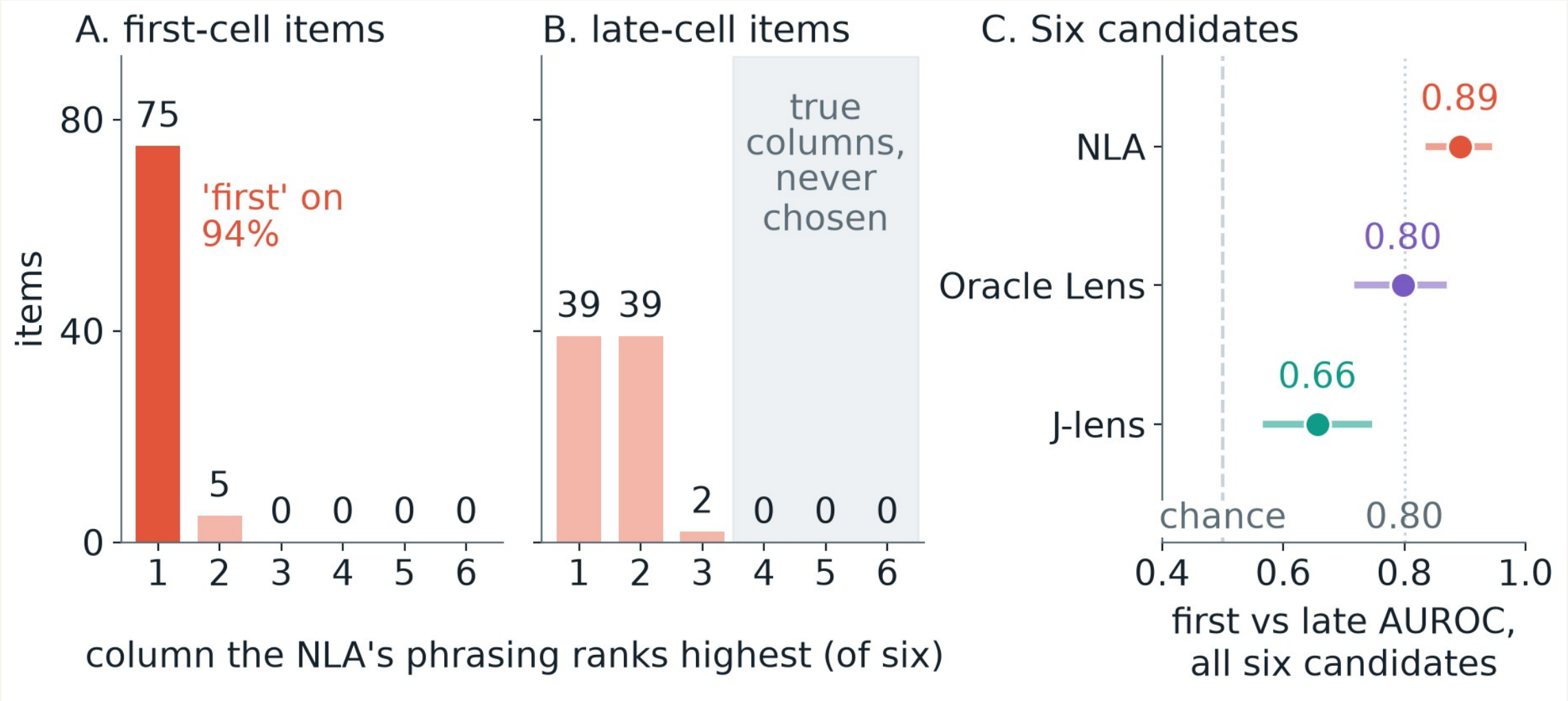
The language control: the NLA states it, Oracle Lens writes in it



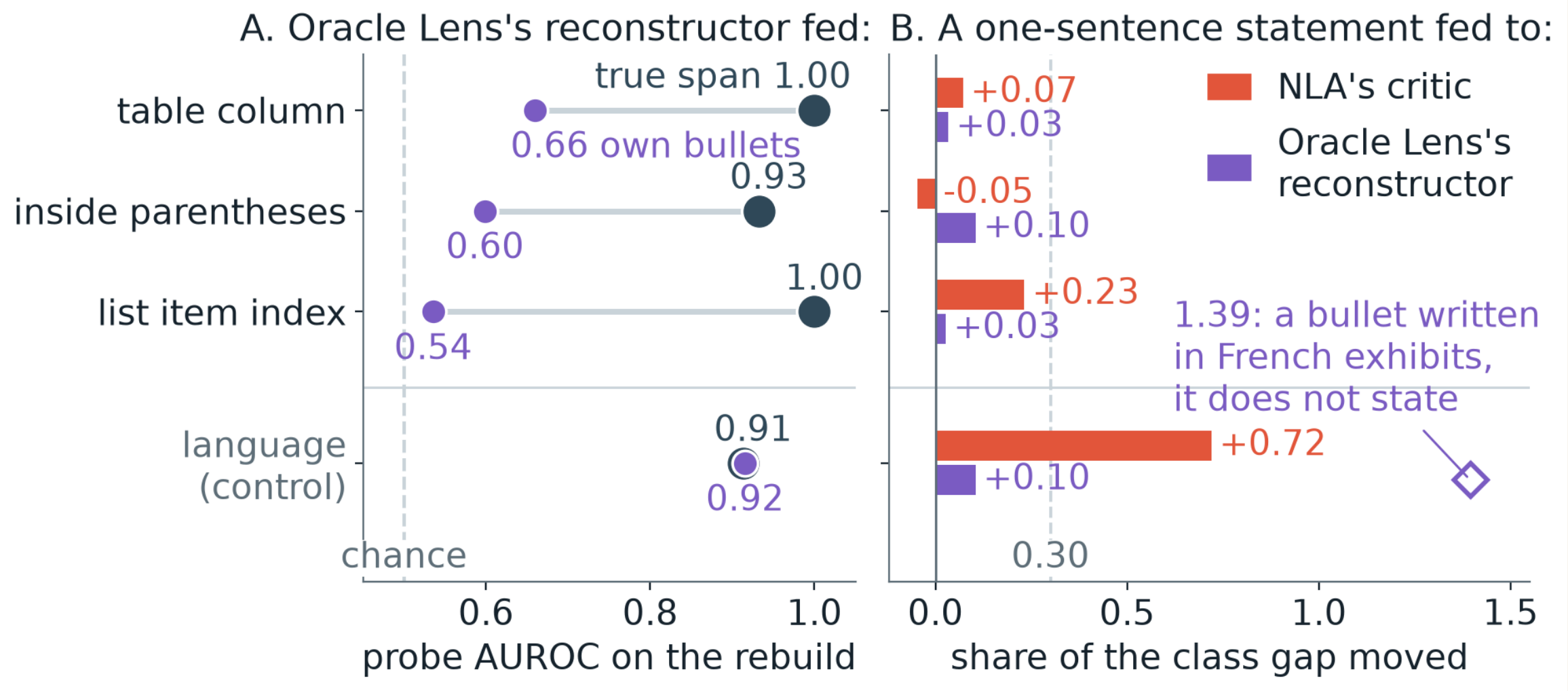
Asked, the NLA detects the position rows; unprompted, Oracle Lens states none



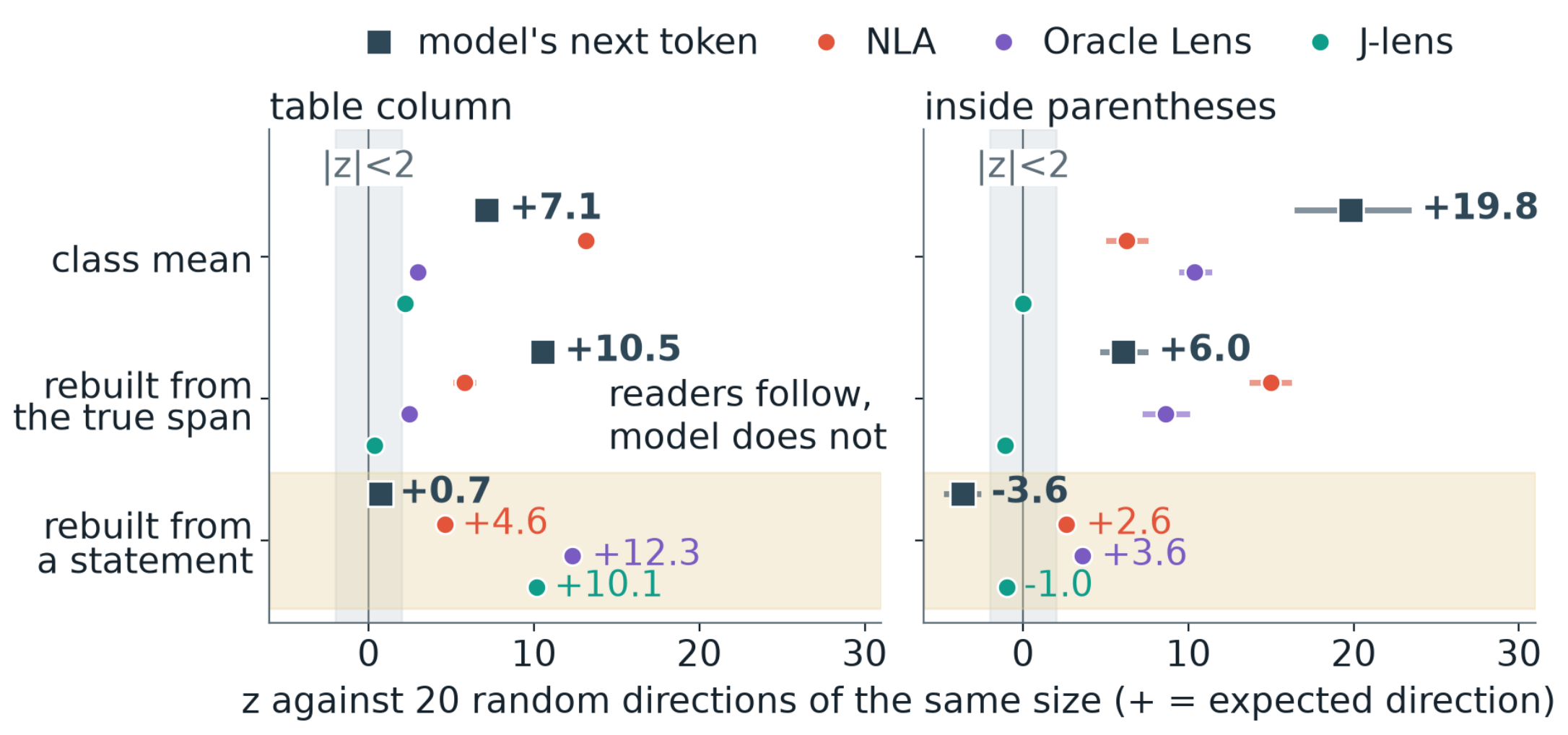
With six columns, the NLA separates first from late, never picks 4 to 6



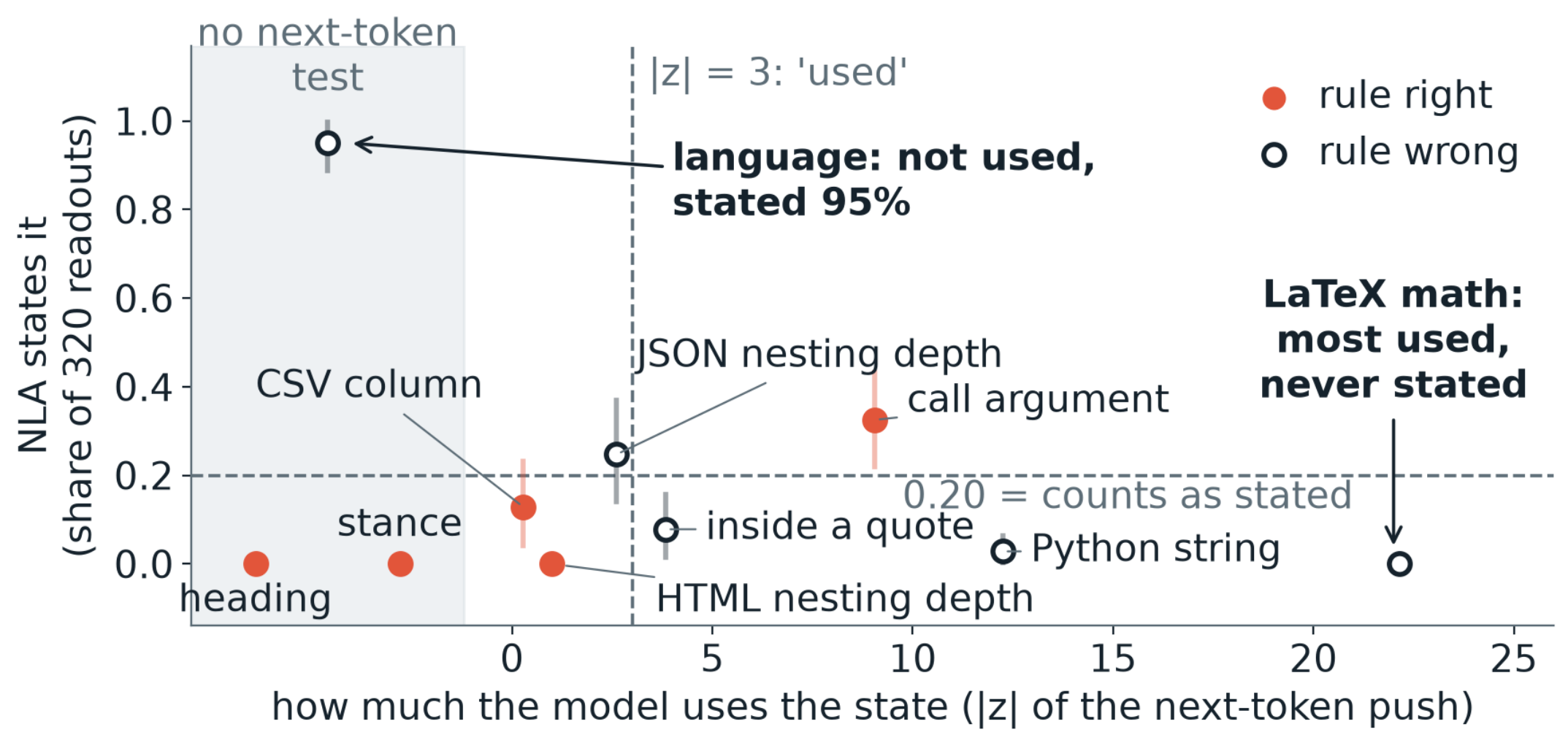
Reconstructors rebuild position from real text, not from a sentence that states it



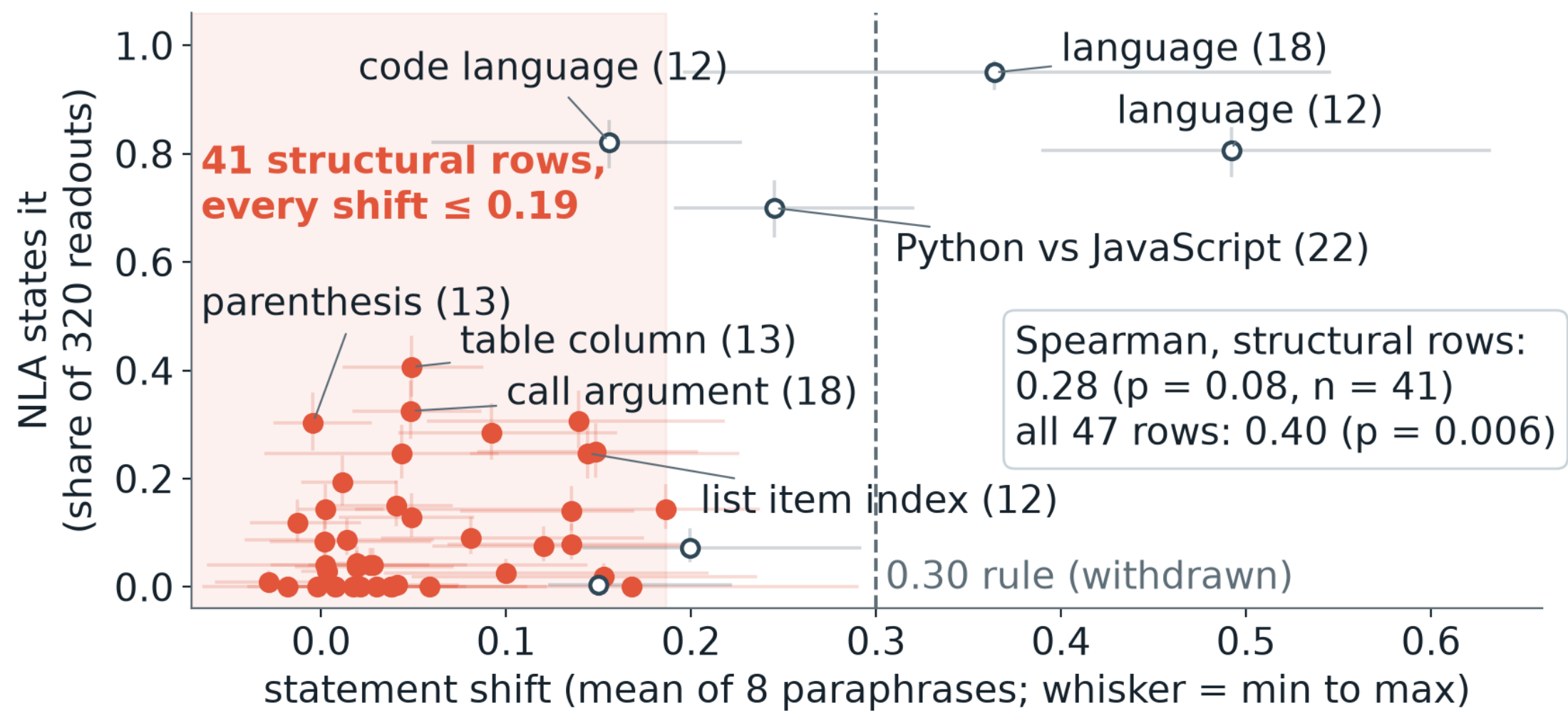
A direction built from a statement moves both readers but not the model



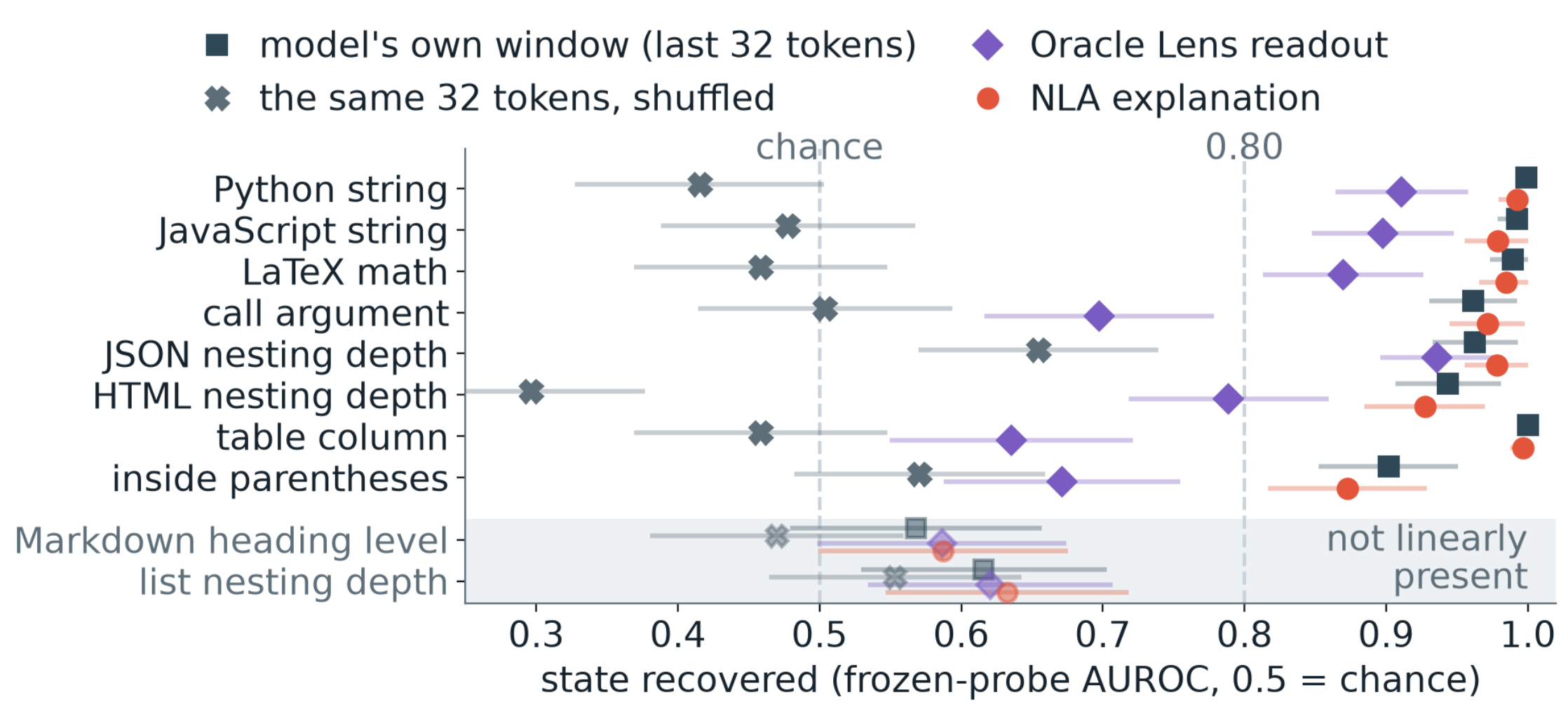
Negative result: exp 18 missed its bar; the NLA does not state what the model uses



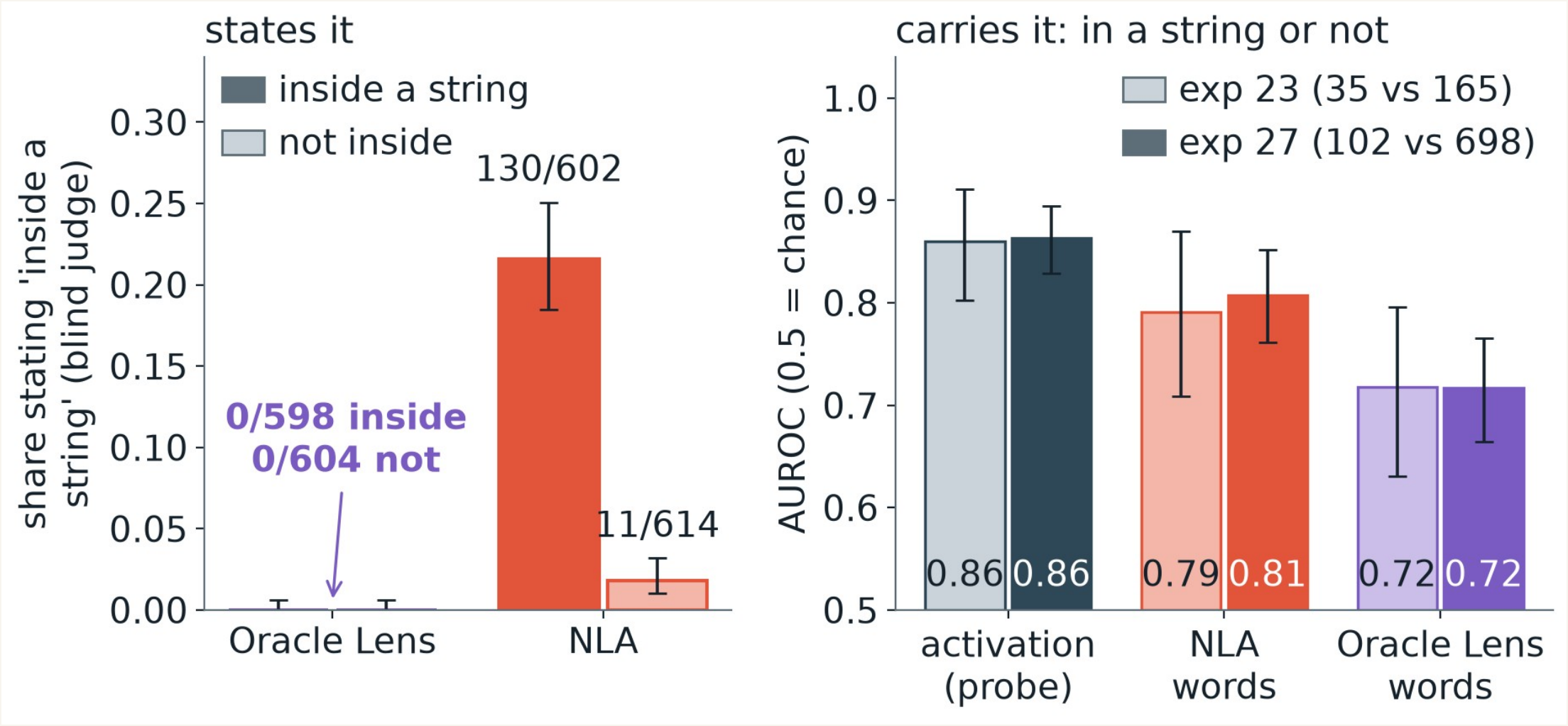
Withdrawn: the statement shift separates language from structure, not rows within structure



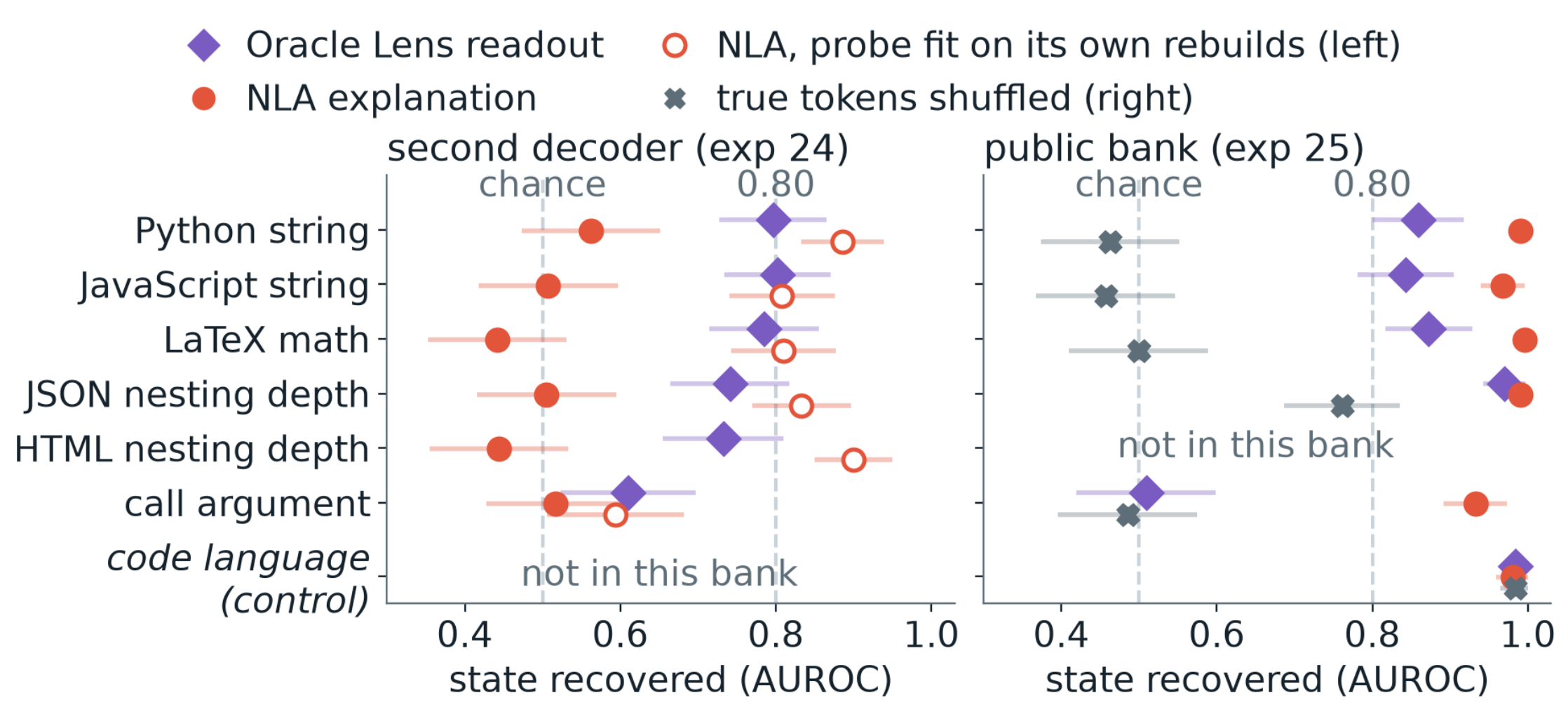
The state is in the model's window and in both readers' words; shuffling removes it



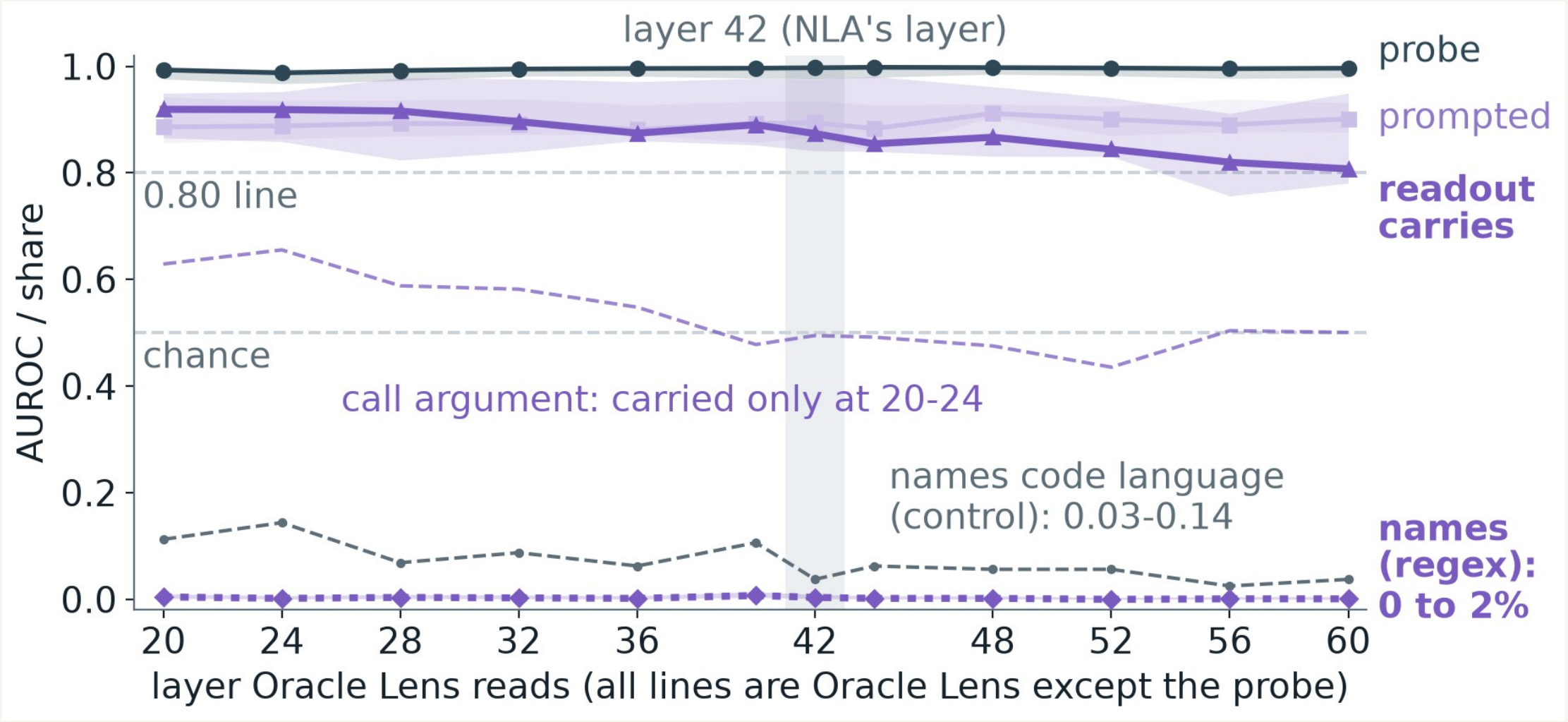
At random read points Oracle Lens never states 'inside a string', yet its words carry it



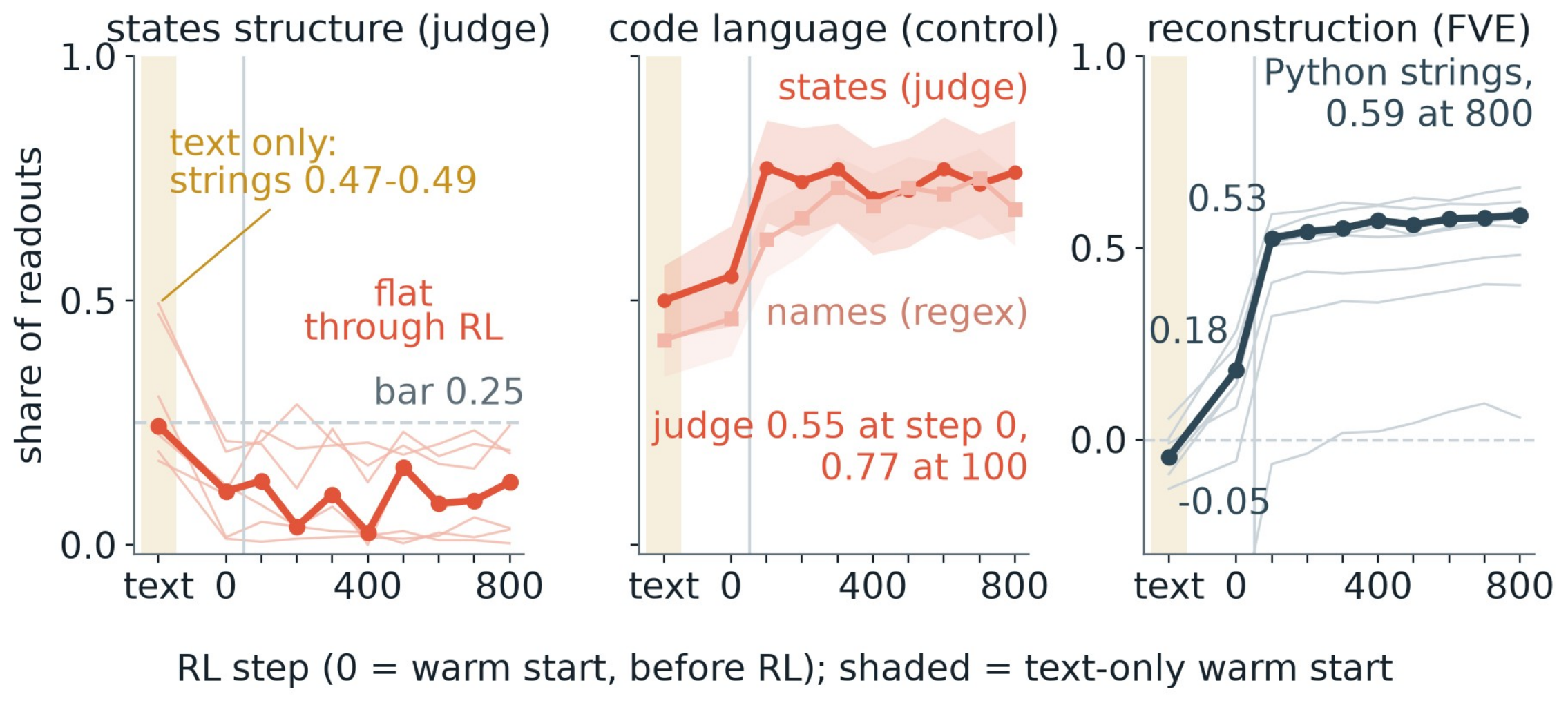
Two checks agree: Oracle Lens's words carry the state



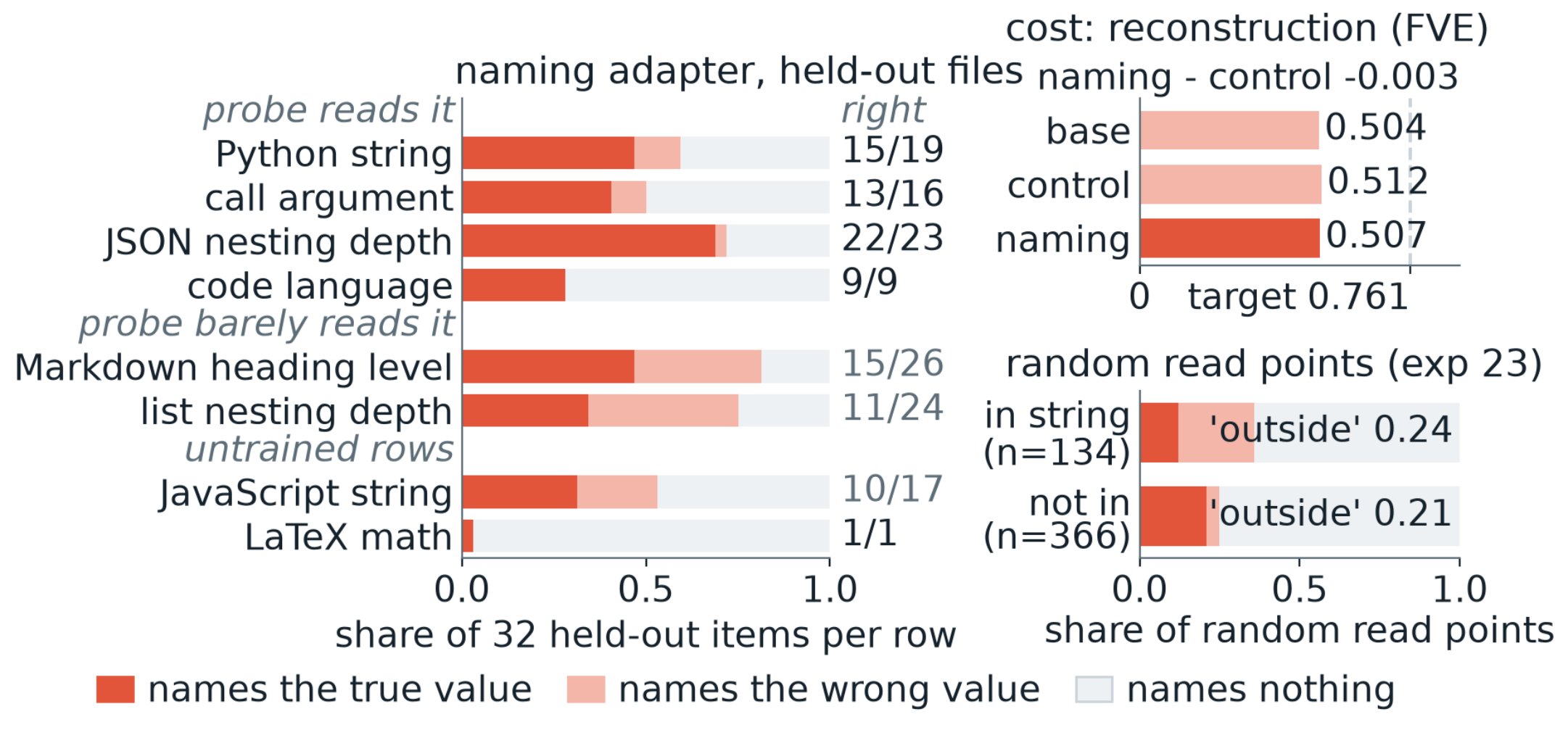
Oracle Lens carries the state and never names it, at every layer from 20 to 60



Along RL, reconstruction and code-language naming rise; structural naming stays flat



Naming costs the reward nothing and is right where readable, but does not generalise



We dropped six ideas during the sprint, each on a logged date

phase		six ideas dropped (log date)	
5-6 Oct	benchmark can't test reading		
6-13 Oct	*J-space pilots (superseded)*		
22-24 Oct	instruments on the activation		
22-26 Oct	position rows: used, not stated		
23-25 Oct	mechanism: span vs statement		
26-27 Oct	pre-registered prediction tests	✗ **26 Oct**	'NLA states it when used'
27-28 Oct	readers carry it, don't name it	✗ **27 Oct**	window account · theory day continuation predictor loose naming count
28-30 Oct	layers, training path, adapter	✗ **28 Oct**	0.30 statement rule

Suraj notes/decisions.md · Jarus77/mats-nla-workspace @ 4d43ecb2